AAE 454 Design of Aerospace Structures
3 credits

1 Course Information/Description

The goal of this course is to equip students with the fundamental principles and knowledge for designing aerospace structural parts based on operational requirements. The course covers design criteria for aerospace structures, operational requirements for major components, finite element analysis (FEA), structural analysis using simulation tools, and damage tolerance analysis.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:

There is not dedicated textbook and class notes will be available through Brightspace Learning Management System. Following reference books will be used.

- Niu, M. C.-Y., Airframe Structural Design, Practical Design Information and Data on Aircraft Structures, Hong Kong Conmilit Press Ltd., Second Edition (4th published), 2011.
- E. F. Bruhn, Analysis & Design of Flight Vehicle Structures, Jacobs Publishing, Inc., 10585 N. Meridian Street, Site 220, Indianapolis, IN 46290. Telephone: 317-844-9400.
- Raymer, D. P., Aircraft Design: A Conceptual Approach. AIAA, Inc., 6th edition, 2018.

3 Assignments

The course has homeworks, team assignments and individual assignments. You will form a team for your team assignments. Thus, we have three types of assignments:

3.1 Weekly Assignments

Weekly homework will be given under this category of assignments. These will be based on weekly topics discussed in the class.

3.2 Individual Term Paper

We will have an individual term paper. The goal of this assignment is to learn about ongoing research and development related to the design of aerospace structures and materials, formal criteria for

assessing the maturity of new technology for design applications, and to gain experience with preparing technical documents.

3.3 Individual Final Project

The aim of this course project is to allow you to investigate a specific topic that you find interesting and is related to aerospace structural design. This will give you an opportunity to learn how to solve an engineering problem using analysis methods/approaches discussed in the class.

4 Course Learning Objectives

On completing this course, the student shall be able to:

- Identify the types of aerospace structures and their operational peculiarities, such as speed, temperature, vibration, maneuvers, performance, etc.
- Determine design constraints, failure criteria, and manufacturing issues.
- Perform material selection, stress and vibration analysis as well as damage tolerance analysis.
- Apply finite element analysis (FEA) for structural analysis.
- Apply design guidelines for major structural components of an aerospace vehicle.

5 Evaluation

Students will be evaluated using following metric for evaluation

Assignments and Exams	**Points/Weights**
Homework	30
Term paper	10
Midterm Exam	30
Final Project	30
	Total: 100

Your grade is based on your performance in the homework, midterm exam, term paper and final project. Assigned letter grades are as follows:

Percentage	Letter grade	Percentage	Letter grade	Percentage	Letter grade
96.67 – 100	A+	80 – 83.32	B-	63.33 – 66.66	D
93.33 – 96.66	A	76.67 – 79.99	C+	60 – 63.32	D-
90 – 93.32	A-	73.33 – 76.66	C	0 – 59.99	F
86.67 – 89.99	B+	70 – 73.32	C-		
83.33 – 86.66	B	66.67 – 69.99	D+		